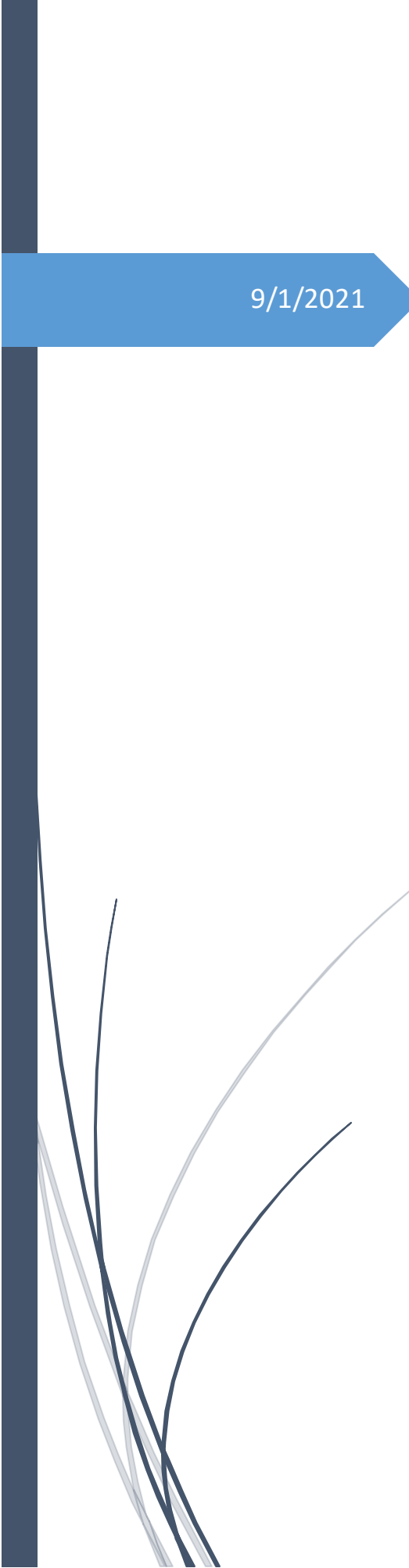


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Introduction to CNET ERP

Chapter 2

Business Management Solutions

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1 Application software

An application program (app or application for short) is a computer program designed to carry out a specific task other than one relating to the operation of the computer itself, typically to be used by end-users. Examples of an application include a word processor and a media player. The collective noun refers to all applications collectively. The other principal classifications of software are system software, relating to the operation of the computer, and utility software ("utilities").

Applications may be bundled with the computer and its system software or published separately and may be coded as proprietary, open-source, or projects. Apps built for mobile platforms are called mobile apps.

1.1 Functions of Application Software

Application software programs are created to facilitate a variety of functions, including but not limited to:

- managing information
- manipulating data
- constructing visuals
- coordinating resources
- calculating figures

1.2 Types of Application Software

Utility software

This could include firewall utilities and antivirus applications, as well as other utilities like zipping or unzipping utilities or disk defragmenting tools, or anything else that an end user can operate as a utility.

Integrated software

Software that does more than one thing, or includes different bundled applications. Here's where your traditional Microsoft Office suite belongs. Another example is a set of database applications bundled together to do different things to data assets.

Specific application software

This would be a single application developed for one defined purpose that is not a utility. Here's where you can separate all of those standalone applications into different categories like games, word processors, analytical engines, newsfeeds etc.

Even social media platforms have come to resemble applications, especially on our mobile phone devices, where individual applications are given the nickname “apps.”

So while the term “application software” can be used broadly, it’s an important term in describing the rise of sophisticated computing environments from early mainframes and Von Neumann models.

2 Management Information system (MIS)

A management information system (MIS) is an information system used for decision-making, and for the coordination, control, analysis, and visualization of information in an organization. It is usually designed for specific organization.

2.1 Types of MIS

The following are types of information systems used to create reports, extract data, and assist in the decision making processes of middle and operational level managers.

- Decision support systems (DSS) are computer program applications used by middle and higher management to compile information from a wide range of sources to support problem solving and decision making. A DSS is used mostly for semi-structured and unstructured decision problems.
- Executive information system (EIS) is a reporting tool that provides quick access to summarized reports coming from all company levels and departments such as accounting, human resources and operations.
- Office automation systems (OAS) support communication and productivity in the enterprise by automating workflow and eliminating bottlenecks. OAS may be implemented at any and all levels of management.

2.2 Advantages of MIS

The following are some of the benefits that can be attained using MIS:

- Improve an organization's operational efficiency, add value to existing products, engender innovation and new product development, and help managers make better decisions.

- Companies are able to identify their strengths and weaknesses due to the presence of revenue reports, employee performance records etc. Identifying these aspects can help a company improve its business processes and operations.
- Giving an overall picture of the company.
- Acting as a communication and planning tool.
- The availability of customer data and feedback can help the company to align its business processes according to the needs of its customers. The effective management of customer data can help the company to perform direct marketing and promotion activities.
- MIS can help a company gain a competitive advantage.
- MIS reports can help with decision-making as well as reduce downtime for actionable items.

2.3 Disadvantages of MIS

Expensive

Installing a management information system can be expensive for a company. Information technology—while cheaper today than previous years—can represent a significant expense, especially for larger organizations. These systems may also require ongoing support or upgrade fees, which can represent future fixed cash outflows. Companies must create a budget to pay for these items to ensure the information system stays current with business technology. Attempting to integrate these systems with technology currently in use can also increase expenses.

Maintenance

Companies may need to hire maintenance individuals to help keep an electronic information system running smoothly. These individuals often need experience in computer science functions and other business topics. Not only does this increase labor costs, but it also requires additional training and ongoing education for these individuals. Business technology can change frequently, creating an environment where companies must have trained individuals who can properly maintain computers, websites, servers and other equipment in use by the management information system.

Ineffective

Management information systems have the potential to become ineffective in a company's operations. Gathering unimportant or non-essential information can delay business decisions because managers must request additional input. Spending too much time reprogramming or correcting issues can also increase the

time spent in the decision-making process. Business owners and managers may also need extensive training on new systems, creating a learning curve that will hopefully diminish over time.

3 Enterprise Resource Planning (ERP)

ERP stands for enterprise resource planning, but what does ERP mean? The simplest way to define ERP is to think about all the core processes needed to run a company: finance, manufacturing, HR, supply chain, services, procurement, and others. At its most basic level, ERP integrates these processes into a single system.

ERP systems tie together a multitude of business processes and enable the flow of data between them. By collecting an organization's shared transactional data from multiple sources, ERP systems eliminate data duplication and provide data integrity with a single source of truth.

Today, ERP systems are critical for managing thousands of businesses of all sizes and in all industries. To these companies, ERP is as indispensable as the electricity that keeps the lights on.

3.1 History of ERP

The history of ERP goes back more than 100 years. In 1913, engineer Ford Whitman Harris developed what became known as the economic order quantity (EOQ) model, a paper-based manufacturing system for production scheduling. For decades, EOQ was the standard for manufacturing. Toolmaker Black and Decker changed the game in 1964 when it became the first company to adopt a material requirements planning (MRP) solution that combined EOQ concepts with a mainframe computer.

MRP remained the manufacturing standard until manufacturing resource planning (called MRP II) was developed in 1983. MRP II featured “modules” as a key software architectural component, and integrated core manufacturing components including purchasing, bills of materials, scheduling, and contract management. For the first time, different manufacturing tasks were integrated into a common system. MRP II also provided a compelling vision of how organizations could leverage software to share and integrate enterprise data and boost operational efficiency with better production planning, reduced inventory, and less waste (scrap). As computer technology evolved through the 1970s and 1980s, concepts similar to MRP II were developed to handle business activities beyond manufacturing, incorporating finance, customer relationship management, and human resources data. By 1990, technology analysts had a name for this new category of business management software—enterprise resource planning.

3.2 ERP Fundamentals

ERP systems are designed around a single, defined data structure (schema) that typically has a common database. This helps ensure that the information used across the enterprise is normalized and based on common definitions and user experiences. These core constructs are then interconnected with business processes driven by workflows across business departments (e.g. finance, human resources, engineering, marketing, operations), connecting systems and the people who use them. Simply put, ERP is the vehicle for integrating people, processes, and technologies across a modern enterprise.

For example: consider a company that builds cars by procuring parts and components from multiple suppliers. It could use an ERP system to track the requisition and purchase of these goods and ensure that each component across the entire procure-to-pay process uses uniform and clean data connected to enterprise workflows, business processes, reporting, and analytics. When ERP is properly deployed at this automotive manufacturing company, a component, for example, “front brake pads,” is uniformly identified by part name, size, material, source, lot number, supplier part number, serial number, cost, and specification, along with a plethora of other descriptive and data-driven items. Since data is the lifeblood of every modern company, ERP makes it easier to collect, organize, analyze, and distribute this information to every individual and system that needs it to best fulfill their role and responsibility.

ERP also ensures that these data fields and attributes roll up to the correct account in the company’s general ledger so that all costs are properly tracked and represented. If the front brake pads were called “front brakes” in one software system (or maybe a set of spreadsheets), “brake pads” in another, and “front pads” in a third, it would be tough for the automotive manufacturing company to figure out how much is spent annually on front brake pads, and whether it should switch suppliers or negotiate for better pricing.

A key ERP principle is the central collection of data for wide distribution. Instead of several standalone databases with an endless inventory of disconnected spreadsheets, ERP systems bring order to chaos so that all users—from the CEO to accounts payable clerks—can create, store, and use the same data derived through common processes. With a secure and centralized data repository, everyone in the organization can be confident that data is correct, up-to-date, and complete. Data integrity is assured for every task performed throughout the organization, from a quarterly financial statement to a single outstanding receivables report, without relying on error-prone spreadsheets.

3.3 The business value of ERP

It's impossible to ignore the impact of ERP in today's business world. As enterprise data and processes are corralled into ERP systems, businesses can align separate departments and improve workflows, resulting in significant bottom-line savings. Examples of specific business benefits include:

- Improved business insight from real-time information generated by reports
- Lower operational costs through streamlined business processes and best practices
- Enhanced collaboration from users sharing data in contracts, requisitions, and purchase orders
- Improved efficiency through a common user experience across many business functions and well-defined business processes
- Consistent infrastructure from the back office to the front office, with all business activities having the same look and feel
- Higher user-adoption rates from a common user experience and design
- Reduced risk through improved data integrity and financial controls
- Lower management and operational costs through uniform and integrated systems

4 ERP Systems

There are a lot of ERP systems developed by different software development companies. In this section we will discuss some of the widely used ERP systems.

SAP

In general, SAP solutions can be deployed in a public or private cloud, or in a hybrid environment. Many of these solutions have embedded AI and machine learning. While many of SAP's solutions can take a relatively long time to implement, the vendor does offer out-of-the-box processes built specifically for an organization's needs. These organizations are typically enterprise-sized and midmarket companies.

Oracle

Oracle provides a set of core applications and industry-specific applications that can be leveraged in basic and advanced ways. The vendor also provides a comprehensive set of cloud options, including public cloud, cloud at customer and on-premises. In 2016, Oracle acquired NetSuite, an application that was born in the cloud and provides solutions for businesses of all sizes and industries.

Microsoft Dynamics

Microsoft offers robust products for small and mid-sized organizations, and is becoming a growing force, looking towards larger organizations as well. According to the report, Microsoft "has a strong ecosystem of ISV (Independent Software Vendors) channel partners providing 'last mile' functionality."

Its Dynamics 365 Business Central, for instance, offers the following features, according to the report:

- Targeted for the small and midmarket space.
- Recently rebuilt in Azure technology, making it a pure Software as a Service (SaaS) model with other deployment options.
- Based on best practices of Net Asset Value (NAV), GP and SL.

Infor (SUN)

Infor is an emerging solution for the enterprise, offering a set of applications specific to particular organizations. The Infor M3 can offer help for after-sales service providers, distributors and a variety of manufacturing verticals.

Infor cloud suite Distribution can allow wholesale distributors to "detect trends and monitor inventory demand," as well as offer organizations a way to "maintain a lean supply chain and implement new digital strategies."

And Infor cloud suite Industrial which is ideal for the manufacturing industry, offers an "open source technology stack, which allows for more flexibility to refine manufacturing processes" and is "designed with features for repetitive manufacturing and capabilities to simplify implementation."

IFS

Crafted for industries such as aerospace and defense, utilities and resources, construction and infrastructure, manufacturing and service industries, IFS has robust end-to-end functionality and is well-suited for large consumer packaged goods organizations with a global footprint.

IFS Applications is a cloud-based enterprise resource planning (ERP) solution that helps businesses to integrate data and processes across multiple departments and locations. It offers various modules to users for asset management, project management, supply chain management and more. IFS Applications helps users to manage end-to-end asset lifecycle within a large enterprise. The solution also offers users capabilities for financial, human resource, document management, CRM business intelligence and reporting operations. The solution holds its applications in a wide range of industries including, construction, engineering, oil and gas drilling, aerospace and manufacturing.

CNET ERP

CNET ERP is a locally developed ERP system by CNET Software Technologies Plc. CNET provides a set of core applications and industry-specific applications that can be deployed as single POS or as advanced integrated set of applications. The system can be deployed at a single customer premises or at multiple locations but in an integrated manner through a cloud service. Currently, the system contains 12 modules and 11 POSs in an integrated manner. The other special nature of the CNET ERP is its integration with different hardware devices.